

CODING

Python	4+ years
C/C++	1 year
Git	2+ years
LaTeX	4+ years

TOOLS

MATLAB/Simulink	4+ years
ROS/ROS2	2+ years
Solidworks	2+ years
Mjlab/MuJoCo	6 months

LANGUAGES

Italian	native
English	C1
German	B1+

CONTACT

Vienna, Austria
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Emilio Gigante
Portfolio

OTHER INFO

- Driving license: Category B
- Basketball: competitive in high school, amateur now

Emilio Gigante

Research assistant in robotics

EXPERIENCE

Project assistant — Apr. 2026 - Present
ACIN - TU Wien / AIT

Developing a deep reinforcement-learning controller for grasping with an automated forestry crane modelled as an underactuated robotic manipulator. Using Mjlab for training and MuJoCo for simulation.

EDUCATION

MSc in Robotics and Automation Engineering — 2022 - 2025
University of Pisa

Thesis: Investigated small language models for on-board AUV mission management, including failure handling and model-confidence evaluation. Integrated the approach into a ROS-based AUV control system using behavior trees, with C++ for ROS components and Python/PyTorch for the language-model component.

BSc in Mechanical Engineering — 2017 - 2022
University of Pisa

Thesis: Analysis of a gravitational accumulator for renewable energy storage.

PROJECTS

Navigation Project — 2024
University of Pisa

Developed an integrated navigation system, that was used to estimate a simulated trajectory of a marine vehicle. MATLAB and Simulink were used to generate the trajectory and to create simulated extended Kalman filters and navigation sensors as accelerometers, gyroscopes, magnetometers and UWB.

Underwater Obstacle Avoidance — 2024
University of Pisa

Contributed to an AUV mission in a real lake with simulated obstacles. Worked on sensor-data processing and environmental-map updates to support path planning, using ROS and Python.

Real-Time Systems Project — 2024
University of Pisa

Developed a real-time 8-ball pool game with a graphic interface for one or two users to play a full game using the mouse and the keyboard. It was written in C and with a higher level library built using the p-thread library.